

Inverse Estimation of Reaction-Rate Coefficients in Oxygen Plasmas using Machine Learning

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1 Summary

Reaction-rate coefficients are a major source of uncertainty in plasma kinetic models, since their values strongly affect predicted species densities and other plasma parameters. This work studies the inverse estimation of selected reaction-rate coefficients in an oxygen plasma model from final species densities generated with LoKI simulations. The aim is to determine a limited number of reaction-rate coefficients, denoted K_i , from final species densities at two pressures, using fully connected neural networks with different species subsets as inputs.

The results show that accurate coefficient recovery is possible when informative species are used. With unperturbed inputs, several combinations reached Scaled MSE values of the order of 10^{-6} . The best-performing combinations were mainly associated with $O_2(a)$ and often with $O_2(b)$, with the pair $O_2(a) + O_2(b)$ already giving very good results. However, adding noise to the input densities caused the error to increase sharply, showing that measurement uncertainty is a central limitation of the inverse approach.

Ensemble averaging reduced the error compared with a single network model, but did not remove the degradation caused by noisy inputs. The individual-species sensitivity analysis showed that robustness depends on which species measurements the model relies on most strongly, while training with noisy inputs improved performance at larger noise levels but reduced accuracy for unperturbed or weakly perturbed inputs. Overall, the method is promising for inverse coefficient estimation, but practical use requires careful species selection and improved robustness to noisy measurements.

2 Introduction and Objectives

2.1 Reaction-Rate Uncertainty in Plasma Models

Low-temperature plasma models depend on kinetic schemes describing how chemical species are created and destroyed. These schemes consist of elementary reactions, each associated with a reaction-rate coefficient that determines how strongly that reaction contributes to species-density evolution [1, 2]. The predicted plasma composition can therefore be strongly affected by the values assigned to these coefficients. Since reaction-rate coefficients are not always known with the same confidence and may come from experiments, calculations, estimates or databases, they are one of the main sources of uncertainty in plasma modelling [2, 3].

Oxygen plasmas are a useful example of this difficulty. Their chemistry includes neutral and charged species, electronically excited states, wall processes, electron-impact reactions and reactions between heavy species [1]. As a result, many coupled processes must be combined into a single kinetic description and small changes in some rate coefficients can propagate through the mechanism and affect the final species densities. Even with a fixed reaction mechanism, rate-coefficient values remain a limiting factor for accurate and reliable simulations.

This motivates the inverse approach followed in this work. Instead of only studying how uncertain rates affect simulated densities, one can ask whether the densities themselves contain

enough information to recover selected rate coefficients [2]. Building on this idea, this work studies the inverse estimation of chosen reaction-rate coefficients from simulated final species densities. The emphasis is not only on whether neural networks can recover the coefficients under ideal conditions, but also on which species provide useful information and how robust the prediction remains when the input densities are affected by measurement noise.

2.2 Inverse Estimation of Reaction-Rate Coefficients

In a conventional plasma-kinetics calculation, the reaction-rate coefficients are specified as part of the kinetic scheme and the model is solved to obtain the time evolution and final densities of the species [1]. The problem considered here follows the opposite direction, in which starting from the final species densities generated by the kinetic model, the goal is to estimate selected reaction-rate coefficients that could have produced them. This turns the recovery of kinetic parameters into a supervised regression problem, where the plasma densities are used as input features and the reaction-rate coefficients are used as target quantities [2].

This inverse mapping is not necessarily unique, since different sets of reaction-rate coefficients may lead to similar final densities, especially when only one pressure condition or a limited number of species is considered [2]. Using the plasma response at two pressure conditions helps constrain the regression problem, because the input data then contain information about how the species densities change across conditions, rather than only a single final state.

A further difficulty in a real experiment is that the available density information may be incomplete or noisy. Although the kinetic model can provide all species densities, an experimental diagnostic would not necessarily measure all of them with the same accuracy. For this reason, the inverse problem depends not only on the regression method, but also on the choice and reliability of the input species. Studying different species combinations, both with unperturbed and noisy densities, therefore helps identify which plasma quantities are most informative and robust for recovering the selected coefficients.

2.3 Scope and Objectives of this Work

Having defined the inverse problem, this work focuses on estimating selected reaction-rate coefficients from final species densities obtained in simulations. The main case considered is an oxygen plasma model with three target coefficients, using neural networks as regression models. Rather than only testing whether the coefficients can be predicted when all input data are available and exact, this work addresses more practical questions, such as which species are most useful as inputs, how the prediction changes when density measurements are noisy and whether the conclusions remain stable across different model architectures and training seeds.

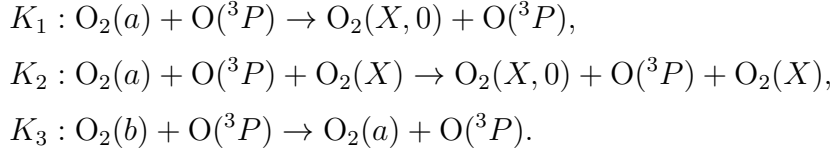
The objectives are therefore to identify informative species combinations for recovering the selected coefficients, evaluate their robustness when noise is added to the input densities, compare single-network and ensemble predictions and examine which individual species measurements are most critical for stable coefficient recovery.

3 Methodology

3.1 Dataset and Inverse Problem Formulation

The main part of this work uses a dataset generated with the LisbOn KInetics simulation tool, LoKI [4], for an oxygen plasma kinetic scheme [1, 2]. Each sample corresponds to a plasma simulation in which selected reaction-rate coefficients are varied and the resulting final heavy-species densities are recorded. These densities are used as inputs to the inverse model, while the varied reaction-rate coefficients are treated as the quantities to be recovered, providing paired examples of plasma compositions and the kinetic parameters that generated them.

In this work, the case study is the estimation of the rate coefficients of three reactions in the oxygen plasma kinetic scheme, denoted here as K_1 , K_2 , and K_3 . These coefficients correspond to the reactions:



The corresponding target vector is therefore $\mathbf{y} = (K_1, K_2, K_3)$. The input vector is built from the final densities of a chosen subset of species evaluated at two pressure conditions. Therefore, if N_s species are selected, each sample has $2N_s$ input features, corresponding to the densities of those species at both pressures. This allows the neural network to use both the absolute density values and the way the plasma composition changes between the two pressure conditions.

The candidate input species considered in the main study are $\text{O}_2(X)$, $\text{O}_2(a)$, $\text{O}_2(b)$, $\text{O}_2(\text{Hz})$, $\text{O}_2^+(X)$, $\text{O}(^3P)$, $\text{O}(^1D)$, O^+ , O^- , $\text{O}_3(X)$ and $\text{O}_3(\text{exc})$ [1, 2]. Since measuring all species predicted by the kinetic model would be difficult in practice, the inverse models are not restricted to using all of them simultaneously. Instead, different subsets are tested as input combinations, making it possible to compare models trained with different available species and to evaluate how much information each combination provides for the inverse prediction.

The data are provided as separate training and test datasets, corresponding to 2000 training samples and 100 independent test samples after grouping the two pressure conditions. During model training, 10% of the training samples are randomly assigned to a validation subset, while the remaining 90% are used to update the neural network weights. Before training, the input densities and target coefficients are scaled using maximum-absolute-value scaling fitted on the training dataset and then applied consistently to the validation and test data, avoiding the use of test-set information during training. After preprocessing, the inverse problem is written as a supervised mapping from the selected density vector $\mathbf{x}$ to the coefficient vector $\mathbf{y} = (K_1, K_2, K_3)$, approximated by the neural network as $\hat{\mathbf{y}} = f_\theta(\mathbf{x})$. The main comparison between species combinations is based on the mean squared error computed in the scaled coefficient space, while predictions are transformed back to the original coefficient scale only when individual coefficient values are inspected directly.

3.2 Neural Network Models and Training Procedure

The inverse mapping is learned using fully connected feed-forward neural networks. For each species combination, the network receives the scaled density vector $\mathbf{x}$ as input and predicts the scaled values of the three reaction-rate coefficients K_1 , K_2 and K_3 . Three architectures are considered: (30, 30), (30, 30, 30) and (50, 50), where each number represents the number of neurons in one hidden layer. The analysis was repeated for all three architectures, with no relevant qualitative changes observed between them. Architecture (30, 30, 30) is used as the representative case in the figures shown in Section 4.

All hidden layers use the hyperbolic tangent activation function. The networks are trained by minimizing the mean squared error between the predicted and true scaled coefficients, using the Adam optimizer with a learning rate of 10^{-4} and a batch size of 16, as these are standard choices for neural network regression and are consistent with previous work on this inverse problem [2]. Each model is trained for a maximum of 5000 epochs, with early stopping applied when the validation loss does not improve for 100 consecutive epochs. This prevents unnecessarily long training and reduces the risk of overfitting to the training subset.

For each architecture and species combination, the training procedure is repeated for 20 different random seeds. These repeated trainings account for the variability introduced by random weight initialization and by the random training-validation split. Results obtained from a single neural network correspond to one independently trained model, while the combination of the independently trained models into an ensemble is described separately in Section 3.3.

3.3 Evaluation Procedures

To compare the information provided by different species, the training and evaluation procedure is repeated for several input combinations. For each combination, only the corresponding species densities are selected from the full input vector, while the target vector remains fixed as $\mathbf{y} = (K_1, K_2, K_3)$. The comparison is performed separately for each network architecture and the performance is evaluated on the independent test set using the Mean Squared Error (MSE) computed in the scaled coefficient space:

$$\text{MSE}_{\text{scaled}} = \frac{1}{3N} \sum_{i=1}^N \sum_{j=1}^3 (\hat{y}_{ij}^{\text{scaled}} - y_{ij}^{\text{scaled}})^2, \quad (1)$$

where N is the number of test samples and y_{ij}^{scaled} and $\hat{y}_{ij}^{\text{scaled}}$ are the true and predicted scaled values of coefficient j for sample i . For each species combination and architecture, the reported performance is obtained by averaging this test-set MSE over the 20 independently trained models, with the corresponding standard deviation used as an indication of training variability. The comparison is therefore interpreted mainly as a way to identify groups of informative combinations, rather than as an absolute ordering.

To evaluate robustness to measurement uncertainty, noise is added to the species densities

before prediction. For an unperturbed density value $x_{i,k}$, the noisy value is written as:

$$x_{i,k}^{\text{noisy}} = x_{i,k} (1 + \epsilon_{i,k}), \quad \epsilon_{i,k} \sim \mathcal{N}(0, \sigma_n^2),$$

where σ_n is the relative noise level. The tested values are 0%, 0.5%, 1%, 2%, 5% and 10%, with the 0% case corresponding to the unperturbed test set. If a perturbation produces a negative density value, the noise is resampled so that the perturbed input remains physically meaningful. For each noise level, 20 perturbed versions of the same test set are generated. Unless stated otherwise, the models are not retrained with noisy data. Only the test inputs are perturbed and the scaled MSE is aggregated over the training seeds and noise realizations.

In addition to the individual neural network models, an ensemble prediction is used to reduce the dependence on a single training seed. For each architecture and species combination, the ensemble is built from the 20 independently trained models. Each model predicts the scaled coefficient vector and the ensemble prediction is obtained by averaging these predicted vectors before computing the error. Thus, the ensemble is treated as a single combined predictor, rather than as an average of the individual model errors.

An individual-species noise sensitivity analysis is also performed. For a given species combination, noise is applied to one selected species at a time, considering its density values at the two pressure conditions, while the remaining input species are kept unperturbed. The same noise levels and multiplicative noise model are used. The sensitivity of each species is assessed from the increase in scaled MSE relative to the unperturbed case, with larger increases indicating that the prediction depends more strongly on the accuracy of that density measurement.

Finally, a second set of neural networks is trained using noisy input densities as an additional robustness test. This procedure is applied only to the representative species combinations and to the architecture (30, 30, 30). During training, each input sample can either remain unperturbed or be perturbed using the same multiplicative noise model. The noise level is randomly selected from 0%, 0.5%, 1%, 2%, 5% and 10%. The unperturbed case is selected with probability 25% while each of the five noisy cases is selected with probability 15%. Only the input densities are perturbed during this procedure, while the target reaction-rate coefficients remain unchanged.

4 Results and Discussion

4.1 Species-Combination Performance without Noise

The unperturbed-input comparison provides the reference case for evaluating the information content of different species combinations before introducing measurement noise. Figure 1 shows a fixed set of representative combinations, including best-performing combinations, simpler reduced combinations, a combination from a different regime and a clearly poor comparison case. This representative set is kept throughout the following discussion so that the behaviour under noise can be compared consistently.

A first observation is that the first 5 combinations all lie in the same low-error regime,

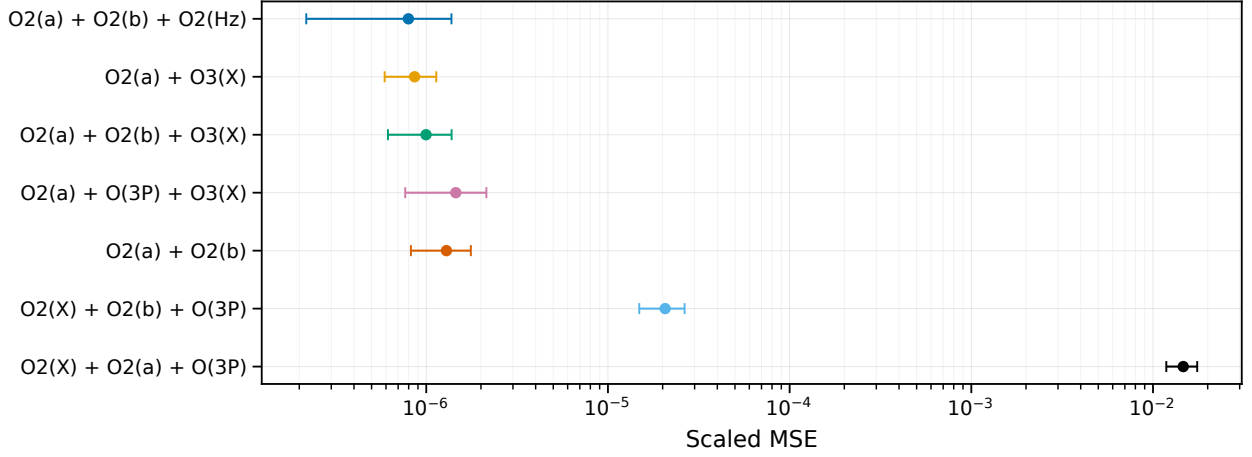


Fig. 1: Representative combinations with unperturbed inputs for the architecture (30, 30, 30). The markers represent the Scaled MSE over the 20 independently trained models, while the error bars represent the corresponding standard deviation. A logarithmic scale is used on the horizontal axis.

with Scaled MSE values of the order of 10^{-6} . Their error bars overlap, so the small differences between these combinations should not be interpreted as defining a single uniquely optimal choice. Instead, these results indicate the existence of a family of informative combinations, dominated by the presence of $O_2(a)$ and often complemented by $O_2(b)$.

The inclusion of the combination $O_2(a) + O(3P) + O_3(X)$ is particularly useful because it shows that a good result can still be obtained without the pair $O_2(a) + O_2(b)$ appearing together. This suggests that the useful information is not tied to one single species pair, but can also be recovered from other combinations when the selected species remain sufficiently informative. At the same time, the good performance of the simpler pair $O_2(a) + O_2(b)$ shows that adding a third species does not necessarily improve the prediction.

The two remaining combinations represent worse regimes. The combination $O_2(X) + O_2(b) + O(3P)$ gives an error around one order of magnitude larger than the low-error group, placing it in an intermediate regime. More importantly, $O_2(X) + O_2(a) + O(3P)$ performs much worse, with a mean scaled MSE of the order of 10^{-2} . This last case is useful because it shows that the presence of $O_2(a)$ alone is not enough to guarantee a good inverse prediction. What matters is the information provided by the full combination of selected species, not only the inclusion of one individually important density.

4.2 Species-Combination Performance under Noisy Measurements

The same representative combinations are tested using noisy input measurements. The noise levels considered are 0%, 0.5%, 1%, 2%, 5% and 10%, where 0% corresponds to the unperturbed reference discussed in Section 4.1. This comparison measures the Scaled MSE under noisy conditions and follows how the error evolves as the noise level increases.

Figure 2 shows that all combinations deteriorate when noise is added to the input densities. The increase is particularly sharp between the unperturbed case and the first noisy case, showing that even small perturbations can strongly affect the inverse prediction. Therefore, unperturbed

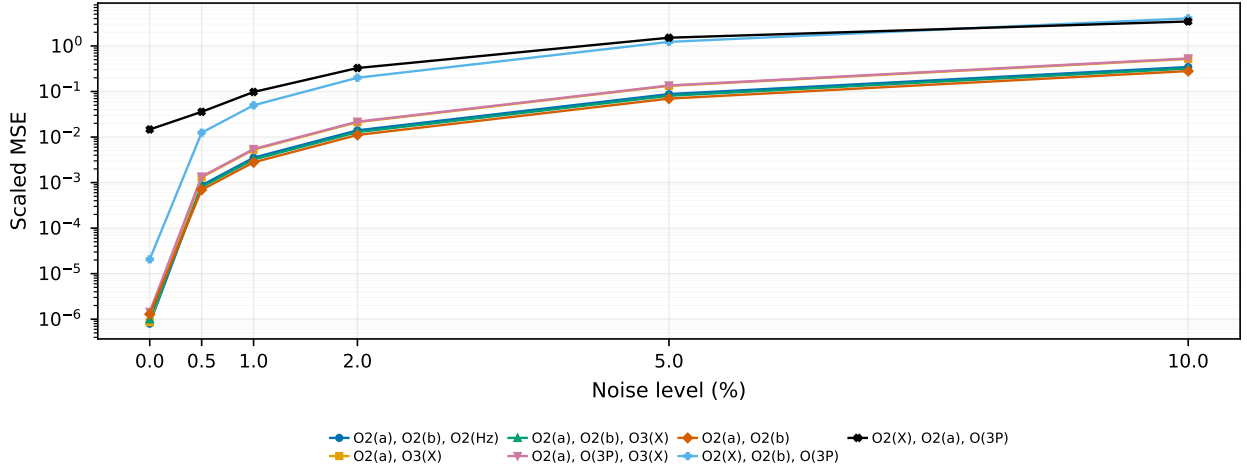


Fig. 2: Scaled MSE with input-noise level for the representative combinations, using the architecture (30, 30, 30). A logarithmic scale is used on the vertical axis.

performance alone gives only a partial view of the quality of each species combination.

The low-error combinations identified in Section 4.1 remain grouped together as the noise level increases. Their curves stay close to each other and follow similar trends on the logarithmic scale, meaning that they degrade in a comparable way under noisy measurements. By contrast, $O_2(X) + O_2(b) + O(^3P)$ no longer behaves as a clear intermediate case once noise is introduced. Already at 0.5% noise, it moves close to the high-error regime of $O_2(X) + O_2(a) + O(^3P)$.

Overall, the results separate into two practical regimes under noisy measurements: a grouped low-error regime and a high-error regime formed by the two comparison cases. Although the unperturbed predictions can be very accurate, the Scaled MSE increases by several orders of magnitude once noise is introduced. Therefore, species combinations should also be evaluated by their robustness to input-density perturbations.

4.3 Ensemble Performance

The results discussed so far were based on the mean performance of individually trained neural networks. A complementary approach is to combine the predictions of the 20 trained models into a single ensemble predictor, as described in Section 3.3. Figure 3 shows the ensemble behaviour for the representative species combinations and the corresponding percentage improvement relative to the single network model, using the architecture (30, 30, 30).

Figure 3 shows that the ensemble preserves the same overall behaviour observed for the single network model. The low-error combinations remain grouped together, while the two comparison cases stay in a worse regime. Therefore, ensemble averaging does not change the main conclusions about which species combinations are more informative.

The main effect of the ensemble is to reduce the Scaled MSE. For the best-performing combinations, the improvement is modest, remaining of the order of a few percent to around 10% for most noise levels. A notable exception is $O_2(X) + O_2(a) + O(^3P)$, which performs poorly without noise in the single-model analysis but, after ensemble averaging, becomes comparable

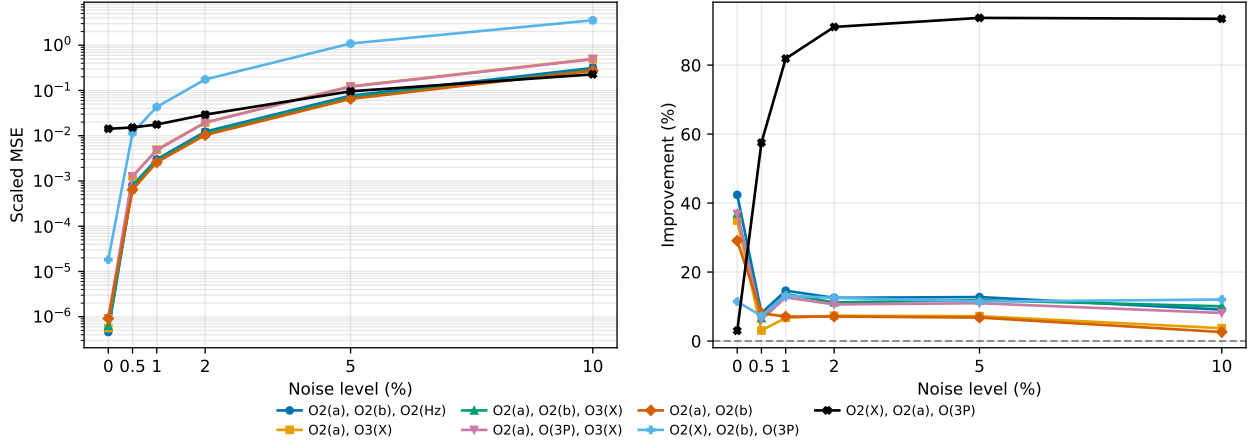


Fig. 3: Ensemble Scaled MSE (left) and percentage improvement relative to the single network model (right) for the representative combinations, using the architecture (30, 30, 30). A logarithmic scale is used on the vertical axis of the left panel.

to the low-error group at larger noise levels. However, the Scaled MSE still increases strongly with noise level, so the ensemble reduces the error but does not remove the sensitivity of the inverse problem to noisy input densities.

4.4 Individual-Species Noise Sensitivity

The previous noise analysis considered the effect of perturbing all selected input species simultaneously. However, this does not indicate which species are mainly responsible for the loss of accuracy in the presence of noise. To examine this, the ensemble model is now evaluated by adding noise only to selected species within a given input combination, while keeping the remaining species unperturbed.

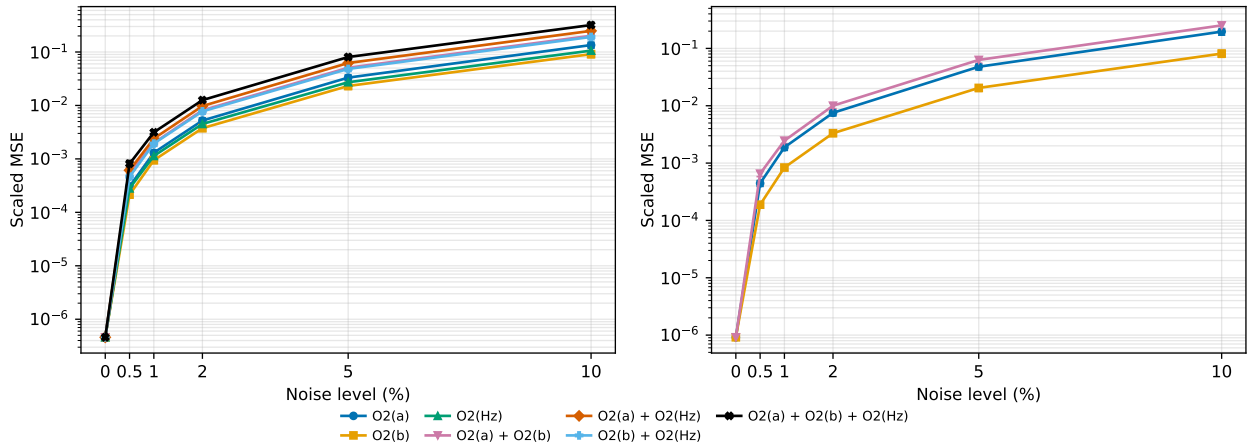


Fig. 4: Ensemble noise sensitivity for selected species perturbations, using the architecture (30, 30, 30). The left panel corresponds to $O_2(a) + O_2(b) + O_2(Hz)$, while the right panel corresponds to $O_2(a) + O_2(b)$. A logarithmic scale is used on the vertical axis.

Figure 4 compares the sensitivity of $O_2(a) + O_2(b)$ and $O_2(a) + O_2(b) + O_2(Hz)$. In both cases, perturbing $O_2(a)$ produces a larger error than perturbing $O_2(b)$, indicating that the inverse prediction depends more strongly on the accuracy of the $O_2(a)$ density measurement.

The three-species case was included to test whether $\text{O}_2(\text{Hz})$ was only a redundant addition to the strong pair $\text{O}_2(\text{a}) + \text{O}_2(\text{b})$. The results show that this is not the case. When noise is applied to $\text{O}_2(\text{Hz})$, the Scaled MSE still increases considerably, meaning that this species is used by the ensemble model. At the same time, perturbing $\text{O}_2(\text{a})$ produces a smaller error in the three-species case than in the two-species case, while the effect of perturbing $\text{O}_2(\text{b})$ remains similar. This suggests that the information provided by $\text{O}_2(\text{Hz})$ partially overlaps with the information extracted from $\text{O}_2(\text{a})$, making the prediction less dependent on $\text{O}_2(\text{a})$ alone. This interpretation should be understood as a behaviour of the trained inverse model, rather than as a direct physical conclusion about the plasma chemistry.

4.5 Training with Noisy Inputs

The previous results show that the inverse prediction is highly sensitive to noise in the input densities. A possible way to improve robustness is to include noisy inputs already during training, instead of training only with unperturbed simulation data. To test this idea, the same representative combinations are evaluated using models trained with noisy input densities and compared with the original ensemble model.

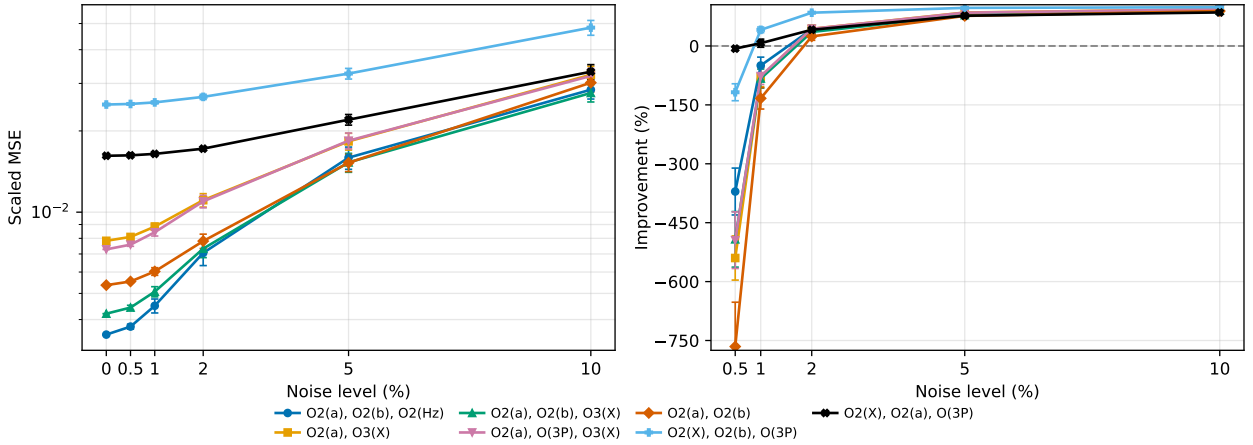


Fig. 5: Ensemble results after training with noisy inputs, using the architecture (30, 30, 30). Left panel shows the Scaled MSE, while the right panel shows the percentage improvement relative to the original ensemble model. Negative values indicate worse performance than the original ensemble.

Figure 5 shows that training with noisy inputs introduces a clear trade-off. At low noise levels, especially close to the unperturbed case, the noisy-trained models perform worse than the original ensemble, as indicated by the negative improvement values in the right panel. Therefore, adding noise during training does not improve the model in all conditions.

At larger noise levels, the behaviour changes. From about 2% noise onward, all combinations show positive improvement relative to the original ensemble, indicating that noisy training makes the inverse prediction more robust when the test inputs are strongly perturbed. However, the left panel still shows that the Scaled MSE increases with noise level and the combinations that were already difficult remain among the highest-error cases. Thus, noisy training improves robustness at larger noise levels, but it does not remove the general sensitivity of the inverse

problem to noisy input densities.

5 Conclusions and Future Work

This work studied the inverse estimation of selected reaction-rate coefficients in an oxygen plasma model using neural networks trained on final species densities. The results show that accurate recovery of the three tested reaction-rate coefficients is possible when informative species are used as inputs. With unperturbed inputs, several combinations reached Scaled MSE values of the order of 10^{-6} . The best results were obtained by a family of combinations mainly involving $O_2(a)$ and often $O_2(b)$, with the pair $O_2(a) + O_2(b)$ already giving very good performance. Therefore, adding species does not necessarily improve the prediction.

The main limitation found in this work is the strong sensitivity of the inverse model to noisy input densities. When noise is introduced, the Scaled MSE increases sharply, so species combinations should be selected not only from their unperturbed performance, but also from their robustness under noisy measurements. The ensemble model reduced the error compared with a single network model, but did not remove this degradation. The additional sensitivity tests showed that robustness depends on which density measurements the model relies on most strongly, with $O_2(a)$ appearing as the most critical input among the combinations studied.

Training with noisy inputs improved robustness at larger noise levels, although at the cost of worse performance for unperturbed or weakly perturbed inputs. Future work should therefore focus on making the inverse model more reliable under realistic measurement conditions. This includes testing the method with experimental data or more realistic diagnostic uncertainties, improving uncertainty quantification for the predicted coefficients and selecting diagnostic species based on both accuracy and robustness. Further improvements could also include more advanced noise-aware training strategies or physically constrained machine-learning models.

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